



COMP 364/464: High-Performance Computing Week 02, Lecture 02



Agenda

1. Course Schedule
2. Project Schedule
3. Policies:
 - a. Readings
 - b. Homeworks
 - c. Projects (3)
 - d. Grades
4. HPC: A Wide Introduction
 - a. Architectures
 - b. Networks
 - c. Topics
5. Projects 1 and 3: Surveys
 - a. What, why, how

Course Schedule

Wk	Date	Topic	Readings	HW	Projects
1	Aug 26	Intro to the class + planning questionnaire	—	—	—
2	Sep 2	Intro to HPC: architectures, networks, topics overview	—	—	Project 1 assigned; Project 3 announced
3	Sep 9	Networks + architectures (cont.) + Linux, finalize	R1	HW1	
4	Sep 16	Parallel programming (part 1)	R2	HW2	Project 1 due; Project 2A assigned
5	Sep 23	Parallel programming (part 2)	R3	HW3	
6	Sep 30	Performance monitoring	R4	HW4	Project 2A due, 2B assigned; Project 3 topic selection due
7	Oct 7	Performance monitoring + midterm review	R5	HW5	
8	Oct 14	Midterm	—	—	Project 3 draft survey due
9	Oct 21	Virtualization in HPC	R6	HW6	Project 2B due
10	Oct 28	Data in HPC + Security	R7	HW7	
11	Nov 4	Fault tolerance	R8	HW8	
12	Nov 11	AI in HPC	R9	HW9	
13	Nov 18	Open — possible guest speaker	R10	HW10	Project work time
14	Nov 25	Thanksgiving break	R11 (opt)	—	
15	Dec 2	Wrap-up / presentations + final review	—	—	Project 3 all parts due

Project Schedule and Contents

Project	Part	Description	Assigned	Due
1	-	Survey of networks & architectures in modern HPC	Wk 2 (Sep 2)	Wk 4 (Sep 16)
2	A	Parallel programming	Wk 4 (Sep 16)	Wk 6 (Sep 30)
2	B	Efficiency analysis of Part A	Wk 6 (Sep 30)	Wk 9 (Oct 21)
3	-	Topic selection (in your area)	Wk 2 (Sep 2)	Wk 6 (Sep 30)
3	A	Survey on your topic (draft Wk 8, final Wk 15)	Wk 2 (Sep 2)	Draft Oct 14 / Final Dec 2
3	B	Programming assignment on your topic	Wk 2 (Sep 2)	Wk 15 (Dec 2)
3	C	Presentation (10 to 20 min, pre-recorded OK; live = extra credit)	Wk 2 (Sep 2)	Wk 15 (Dec 2)

Policy: Readings

- Posted weekly (starting this week)
 - Do them by the following week to follow-along with the lecture successfully
- Technically optional, since I won't be checking, but highly-recommended
- Reminder: get the textbook!
 - Distributed Systems (free PDF): <https://www.distributed-systems.net/index.php/books/ds4/>

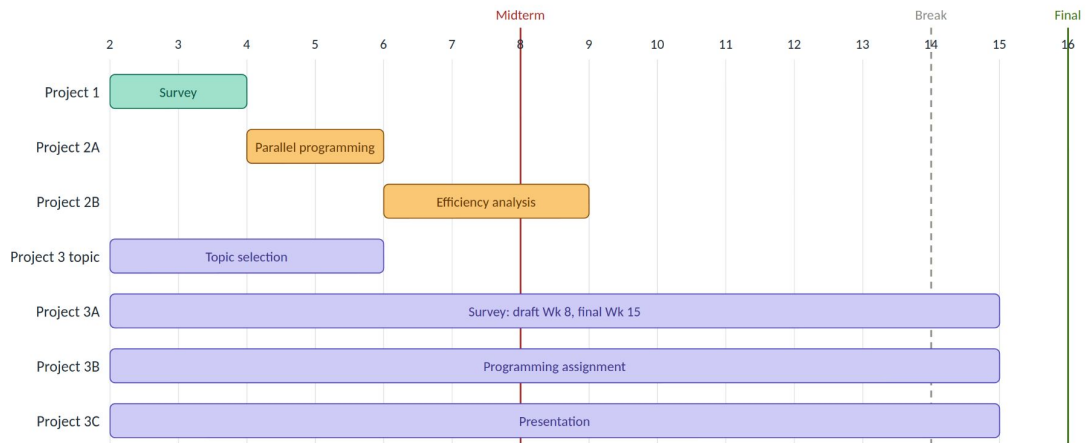
Policy: Homeworks

- Posted weekly alongside the readings
- Entirely optional. These **do not** count towards your grade.
- Solutions posted the following week
 - If you submit a version, I will check it over with you, especially any problem with a subjective answer
 - Please submit on-time -- I will not review late homeworks

Policy: Projects

- Three total: Systems Survey, Programming and Performance, and Capstone
- Project 1: Systems Survey
 - A short research survey of 5-10 machines from the Top500 list
 - Assigned Wk 2, due Wk 4.
- Project 2: Programming and Performance
 - Part A: Write parallel code and get it running correctly across cores/nodes
 - Assigned Wk 4, due Wk 6.
 - Part B: Efficiency analysis. Takes the code from 2A and measures it: profile it, find where the time goes, and analyze how well it scales
 - Assigned Wk 6, due Wk 9.
- Project 3: Capstone
 - Self-selected topic.
 - Four parts:
 - Topic selection: Pick and clear a topic. You are encouraged to pursue your interests! Announced Wk 2, due Wk 6.
 - 3A, Survey: A written research survey on the same topic, with a draft due at the midterm (Wk 8) and the final version due Wk 15.
 - 3B, Programming: A hands-on implementation on chosen topic. Due Wk 15.
 - 3C, Presentation: A brief 10 to 20 minute talk, pre-recorded is fine; presenting live earns extra credit. Due Wk 15.

Project Schedule and Contents



Policy: Grades

- Exams:
 - Midterm: 25%
 - Final: 25%
 - Example exams (for format) and solutions (for expectations) will be provided at the review
 - Depending on curve, might modify this to 20-30 if it's advantageous to all students
- Projects:
 - P1, Systems survey: 10%
 - P2, Programming and Performance: 15%
 - P3, Capstone: 25%
 - Rubrics will be posted

Questions?

10 Minute Break

HPC: A Wide Introduction

1. Systems
2. Architectures
3. Networks
4. Users

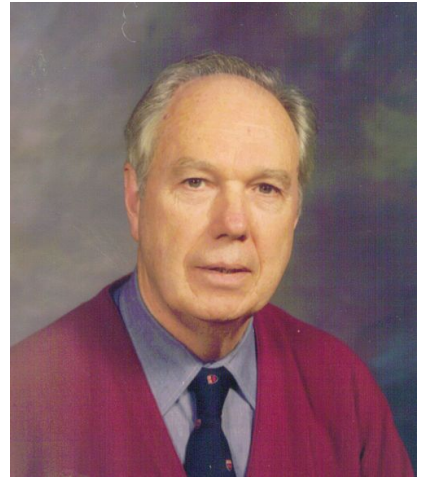
Systems

- Varied over time
 - In the past: special-purpose machines
 - In the 90-00s: general-purpose was key!
 - In the 10s: GPUs, everywhere
 - Now: back to special-purpose
- Top500
 - Ranks according to throughput using HPL, or high-performance Linpac
 - Re-ranked periodically
- Examples:
 - We are going to go through many!

But first, we need a simple taxonomy.

System Taxonomy: Flynn

- Michael J. Flynn
 - PhD, Purdue in 1961
 - Founding chairman, IEEE TCAA
 - Founding chairman, ACM SIGARCH
 - *Got the ACM/IEEE groups to sit in the same room*
- Awards:
 - Harry H. Goode Memorial Award
 - IEEE Computer Society Charles Babbage Award
- Classified systems into a concise taxonomy in 1966
 - Is it perfect?
 - No - but it's the best starting point out there!



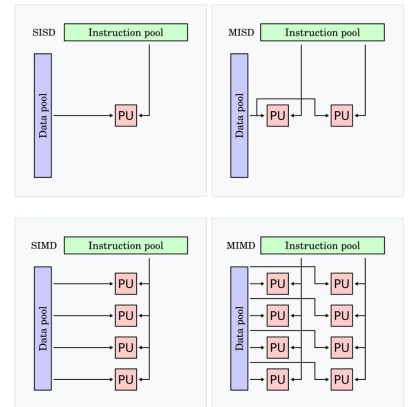
https://en.wikipedia.org/wiki/Flynn%27s_taxonomy

https://en.wikipedia.org/wiki/Michael_J._Flynn

https://en.wikipedia.org/wiki/Duncan's_taxonomy#Shared_memory

System Taxonomy: Flynn

- Broke instruction stream (calculation to run) away from data stream (what to calculate *on*)
 - Single or Multiple in both cases
- SISD
 - Simplest standard computer we can imagine
- SIMD:
 - GPU! Or vector processors, more specifically.
- MISD
 - Not common - streaming computers, fault-tolerance
- MIMD
 - Bog-standard multicore processors



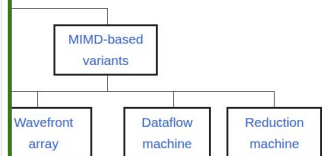
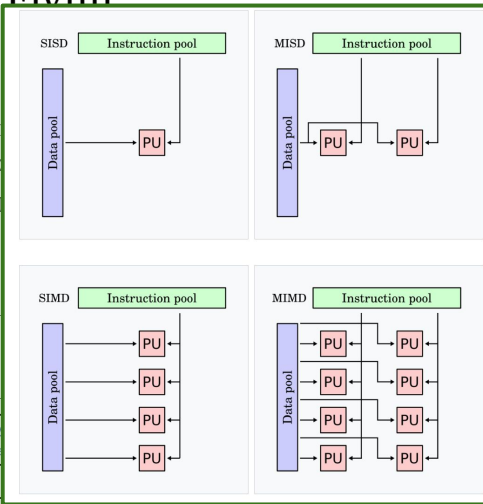
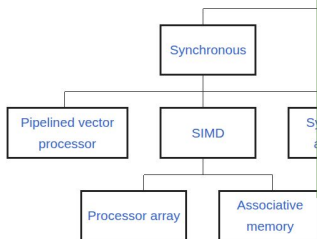
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System Taxonomy: Flynn

- What's missing?
 - Vector pipelines, for o
 - MISD doesn't really ex
- Can go further, Duncan



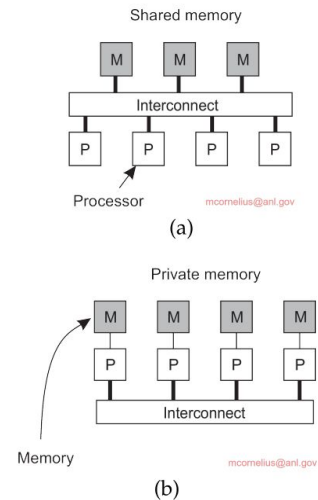
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https://en.wikipedia.org/wiki/Duncan's_taxonomy#Shared_memory

Memory Addressing: Distributed vs Shared

- MIMD: multi-instruction, multi-data
 - What happens when they want to communicate?
- Distributed (each processor is private):
 - Each system has its own memory
 - Pass messages containing requests/data
 - Good: Easy, extendable, safe, debugging is simpler
 - Bad: Slow. So, so slow.
- Shared (processors are linked):
 - Systems share a bank of memory
 - Coordination uses locking/ownership
 - Good: Fast! And all in one place.
 - Bad: Complex, extremely. Debugging can be opaque. Unsafe.
- These days: We do both!



From our text: M. van Steen and A.S. Tanenbaum, Distributed Systems, 4th ed., distributed-systems.net, 2023.

Real Life: Distributed vs Shared

- Distributed memory:
 - Multi-process programming: MPI, OpenMP
 - Multi-processor programming: MPI, distribution frameworks
 - GPU programming: multi-GPU, frameworks
- Shared memory:
 - Threads! Pthreads, OpenMP
 - GPU programming: Native
- More on these (lots and lots more) in a few weeks.
- For now, takeaways:
 - Flynn's Taxonomy
 - Distributed vs shared memory programming
 - Gratitude for our compliance standards, compilers, frameworks, and the wonderful people who maintain them

OpenMP®

MPICH

OpenCL™

AMD
ROCm

NVIDIA

<https://informatcdigital.com/en/what-is-openmp/>

<https://www.mpich.org/>

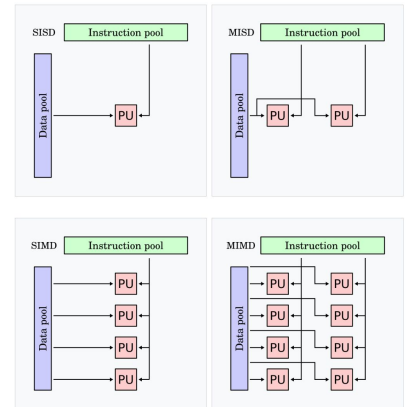
<https://en.wikipedia.org/wiki/OpenCL>

<https://logodix.com/cuda>

<https://www.amd.com/en/developer/resources/rocm-hub.html>

System Taxonomy: Flynn

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https://en.wikipedia.org/wiki/Duncan's_taxonomy#Shared_memory

Back to Systems

- Goal: get a solid feel for where HPC was, where it is now, and the leaps made
- Why: we can't see where we're going if we can't define where we've been
- How: let's get geeky!
 - System-by-System breakdown
 - Year
 - Usage
 - Speed
 - What made the system special
- Then we'll talk about networks

We deserve a breather.

Questions?

10 Minute Break

Back to Systems

- Goal: get a solid feel for where HPC was, where it is now, and the leaps made
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 - Year
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 - Speed
 - What made the system special
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Top500



- How do we rank supercomputers?
- Needed a standard!
- Uses LINPACK (actually HPL, the MPI-distributed version)
 - Dense linear algebra solver
 - Basically, can use to summarize floating point capability with a single number (actually 4)
 - Common scientific task, scalable (massively), can use to push for the highest compute without scaling memory, etc
 - More like a ceiling than an actual speedometer
 - ...Know what, good enough

From Top500 itself:

"This performance does not reflect the overall performance of a given system, as no single number ever can. It does, however, reflect the performance of a dedicated system for solving a dense system of linear equations."

- 1 **LineShine** - LingKun, LX2 304C 1.55GHz, LingQi, Kylin OS, **Shenzhen Cloud Computing Center Co., Ltd.**
- 2 **El Capitan** - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, **HPE**
- 3 **Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, **HPE**
- 4 **Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, **Intel**

<https://www.top500.org/>

CDC 6600 (1964)

Control Data Corporation (Seymour Cray),
Chippewa Falls, Wisconsin

When & where. The first machine widely called a supercomputer. Designed by Seymour Cray. About 100 were sold. Fastest in the world 1964 to 1969.

Architecture. A single 60-bit CPU with 10 parallel functional units. Input/output handled by 10 small peripheral processors. Freon-cooled. Introduced the president for out-of-order execution and is often cited as an early RISC-like design.

Network topology. Not a networked machine. Parallelism came from the 10 units inside one CPU plus 10 peripheral processors around it.

Speed in perspective. About 3 million calculations per second, roughly 3 times faster than the previous record holder (IBM 7030 Stretch). A modern phone is around a million times faster.

Why it mattered. Coined the term supercomputer, established Seymour Cray, and set the template of a purpose-built fast machine. Vector drawing interface with a terminal replaced knobs and tubes. Honestly, just a very neat computer!



Key specs

Speed	~3 MFLOPS (3 million ops/sec)
Processors	One 60-bit CPU, 10 MHz, 10 functional units
Nodes	Single system + 10 peripheral processors
Memory	Up to ~982 KB
Interconnect	None (internal functional units)
Power	~30 kW

https://en.wikipedia.org/wiki/CDC_6600

<https://digibarn.com/collections/systems/crays/cray1/boards/index.html>

<http://ygdes.com/CDC/cdc6600.html>

http://ygdes.com/CDC/DesignOfAComputer_CDC6600.pdf

CDC 6600 (1964)

Control Data Corporation Chippewa Falls, Wisconsin

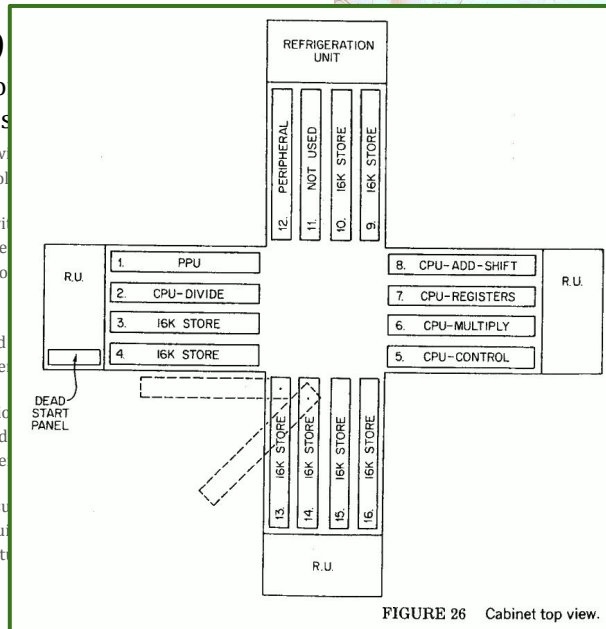
When & where. The first machine was built in 1964 by Seymour Cray. About 100 were sold.

Architecture. A single 60-bit CPU with 10 functional units. Input/output handled by 10 small peripheral processors. Introduced the president for out-of-order execution, an early RISC-like design.

Network topology. Not a networked system. Multiple units inside one CPU plus 10 peripheral processors.

Speed in perspective. About 3 million operations per second, 100 times faster than the previous record. A telephone is around a million times faster.

Why it mattered. Coined the term supercomputer and set the template of a purpose-built machine. With a terminal replaced knobs and toggle switches.



3 million ops/sec)
CPU, 10 MHz, 10 functional units
+ 10 peripheral processors
3
al functional units)

FIGURE 26 Cabinet top view.

https://en.wikipedia.org/wiki/CDC_6600

<https://digibarn.com/collections/systems/crays/cray1/boards/index.html>

<http://ygdes.com/CDC/cdc6600.html>

http://ygdes.com/CDC/DesignOfAComputer_CDC6600.pdf

CDC 6600

Control Data Corp
Chippewa Falls, W

When & where. The first released in 1964 by Seymour Cray. About 10

Architecture. A single 60-bit word. Input/output handled by 12 channels. Introduced the president for an early RISC-like design.

Network topology. Not a shared bus. Units inside one CPU plus 12 peripheral processors.

Speed in perspective. About 10 times faster than the previous CDC 6400. A phone is around a million times faster.

Why it mattered. Coined the term "Cray" and set the template of a personal computer with a terminal replaced by a computer!

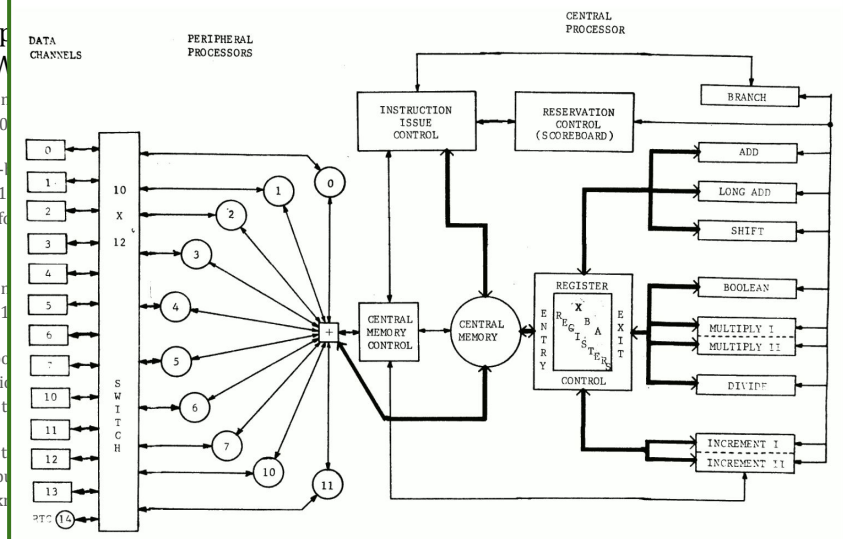


FIGURE 26 Cabinet top view.



onal units
processors

https://en.wikipedia.org/wiki/CDC_6600

<https://digibarn.com/collections/systems/crays/cray1/boards/index.html>

<http://ygdes.com/CDC/cdc6600.html>

http://ygdes.com/CDC/DesignOfAComputer_CDC6600.pdf



Cray-1 (1976)

Cray Research

Serial #00001 at Los Alamos National Laboratory

When & where. The iconic vector supercomputer. Serial 001 went to Los Alamos in 1976; NCAR was the first paying customer. Fastest in the world 1976 to 1982.

Architecture. A single 64-bit vector processor with eight 64-element vector registers and chaining. Built in a C shape so no wire ran more than four feet; the padded base hid the power supplies. Freon cooled.

Network topology. A single-processor, shared-memory machine with no interprocessor network. Its speed came from vector pipelining and chaining, not parallel nodes.

Speed in perspective. 160 million operations per second, about 10 times faster than anything else in 1976. It became the yardstick other machines were measured against.

Why it mattered. The first commercially successful supercomputer; over 80 sold. The machine most people still picture when they hear the word supercomputer.



Key specs	
Speed	160 MFLOPS peak, 80 MHz
Processors	One 64-bit vector processor
Nodes	Single processor, shared memory
Memory	8 MB (1M 64-bit words)
Interconnect	None (single processor)
Power	~115 kW

<https://en.wikipedia.org/wiki/Cray-1>



Cray-1 (1976)

Cray Research
Serial #00001 at LBNL

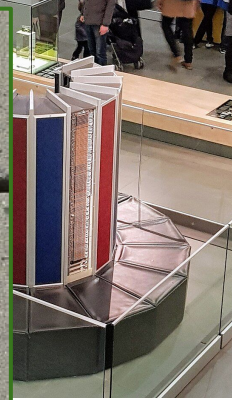
When & where. The icon of supercomputing, built at Los Alamos in 1976; Most powerful machine in the world 1976 to 1985

Architecture. A single 64-element vector register file. Each processor wire ran more than four miles. Freon cooled.

Network topology. A simple bus with no interprocessor communication. Pipelining and chaining.

Speed in perspective. 1.6 GFLOPS, 10 times faster than anything else in the world other machines were measured in MFLOPS.

Why it mattered. The first machine to break the 1 MFLOPS barrier. Over 80 sold. The machine that started the word supercomputer.



1.6 GFLOPS peak, 80 MHz
64-bit vector processor
1 processor, shared memory
1M 64-bit words
1 single processor
W

<https://en.wikipedia.org/wiki/Cray-1>

Connection Machine CM-5

Thinking Machines Corp., 1993

No. 1 machine at Los Alamos National Laboratory

When & where. Topped the very first TOP500 list in June 1993. The No. 1 configuration had 1,024 nodes at Los Alamos. Thinking Machines built five of the top ten machines on that list.

Architecture. A massively parallel MIMD machine: up to 1,024 RISC SPARC nodes, each with vector units and its own memory, communicating by message passing. Designed by Danny Hillis's team.

Network topology. A fat-tree data network (a landmark HPC topology from Charles Leiserson), plus a separate control network for broadcast and synchronization and a diagnostic network.

Speed in perspective. About 60 billion operations per second. Today a single high-end phone or game console blows it out of the water.

Why it mattered. One of the first successful massively parallel machines. Thinking Machines went bankrupt in 1994. Anyone recognize the movie?



Key specs	
Speed	59.7 GFLOPS LINPACK (131 GFLOPS peak)
Processors	1,024 SPARC nodes, 32 MHz, with vector units
Nodes	1,024 nodes, 32 MB each
Memory	~32 GB total
Interconnect	Fat-tree data network
Power	n/a

<https://www.starringthecomputer.com/appearance.html?f=11&c=15>

<https://www.threads.com/@dreamwieber/post/DWWgdcwFLE4/the-connection-machines-saga-is-fascinating-you-can-learn-more-about-it-here>

https://en.wikipedia.org/wiki/Connection_Machine

Connection Machine CM-5

Thinking Machine
No. 1 machine at

When & where. Topper
No. 1 configuration had
Machines built five of t

Architecture. A massiv
SPARC nodes, each wit
communicating by mes
team.

Network topology. A f
topology from Charles
for broadcast and sync

Speed in perspective.
Today a single high-end
water.

Why it mattered. One
machines. Thinking Ma
recognize the movie?



Power

n/a

<https://www.starringthecomputer.com/appearance.html?f=11&c=15>

<https://www.threads.com/@dreamwieber/post/DWWgdcwFLE4/the-connection-machines-saga-is-fascinating-you-can-learn-more-about-it-here>

https://en.wikipedia.org/wiki/Connection_Machine

Intel ASCI Red (1997)

Sandia National Laboratories, Albuquerque
Built by Intel for DOE

When & where. The first machine to break one teraflop, in June 1997. Built for the DOE Accelerated Strategic Computing Initiative. Held No. 1 from 1997 to 2000.

Architecture. A MIMD massively parallel machine of over 9,000 commodity Intel Pentium Pro processors (later upgraded to Pentium II OverDrive), across 104 cabinets and about 2,500 square feet. The first large supercomputer built entirely from off-the-shelf parts.

Network topology. A split-plane mesh, 38 by 32 by 2, partitioned into compute, service, input/output, and system sections.

Speed in perspective. One trillion operations per second, a major milestone. Roughly the double-precision throughput of a mid-range modern GPU (across a large room of PCs in 1997).

Why it mattered. First teraflop machine; proved commodity CPUs could scale to the very top, and launched the ASCI program that funded a generation of US leadership machines.



Key specs

Speed	1.068 TFLOPS LINPACK (first teraflop)
Processors	~9,200 Intel Pentium Pro, 200 MHz
Nodes	~4,500 compute nodes (2 CPUs each)
Memory	~1.2 TB distributed
Interconnect	38 x 32 x 2 mesh
Power	~850 kW

https://en.wikipedia.org/wiki/ASCI_Red

<https://www.tomshardware.com/picturestory/696-supercomputer-evolution-over-22-years.html>

Intel ASCI Red

Sandia National Laboratory
Built by Intel for DOE

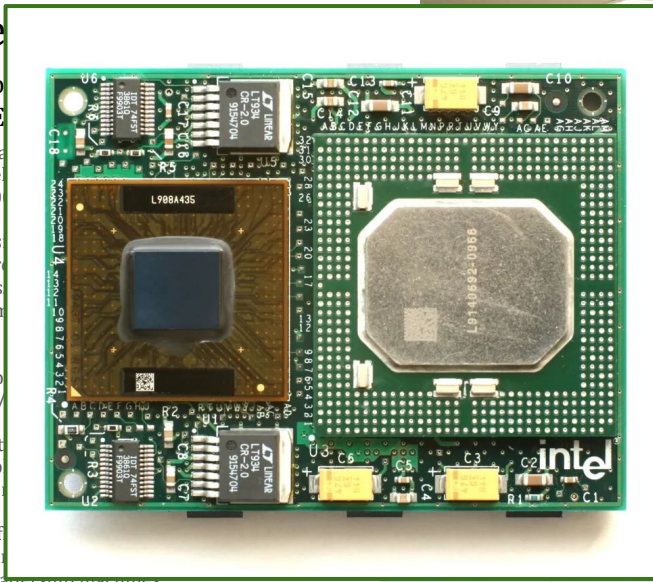
When & where. The first machine was built in 1997. Built for the DOE Accelerator Computing Program. Held No. 1 from 1997 to 2000.

Architecture. A MIMD mass parallel architecture using commodity Intel Pentium Pro (Pentium II OverDrive), across 8,192 nodes. The first large supercomputer to use commodity parts.

Network topology. A split-pipe architecture into compute, service, input/output.

Speed in perspective. One teraflop milestone. Roughly the double of a modern GPU (across a large number of nodes).

Why it mattered. First teraflop machine. Could scale to the very top, and funded a generation of US leadership machines.



INPACK (first teraflop)
Pentium Pro, 200 MHz
8,192 nodes (2 CPUs each)
1,000,000,000,000 operations per second
sh

Power ~850 kW

https://en.wikipedia.org/wiki/ASCI_Red

<https://www.tomshardware.com/picturestory/696-supercomputer-evolution-over-22-years.html>

Earth Simulator (2002)

JAMSTEC, Yokohama, Japan

Built by NEC

When & where. Held No. 1 from June 2002 through mid-2004, across five lists. Housed in a purpose-built hall the size of about four basketball courts.

Architecture. A highly parallel vector machine: 640 shared-memory nodes, each with eight NEC SX-6 vector processors, for 5,120 processors and 10 TB of memory. It reached about 87.5% of peak on LINPACK, unusually high.

Network topology. A single-stage, non-blocking crossbar (the IXS) linking all 640 nodes, about 12.3 GB/s each direction between any two nodes

Speed in perspective. About 36 trillion operations per second, dedicated to modeling the planet's atmosphere and oceans at roughly 10 km resolution. It jumped nearly 5x past the previous No. 1, a massive jump.

Why it mattered. Computenik moment. Shocked the US into renewed supercomputing investment, and a showcase for vector-plus-crossbar design pointed at climate science domain (and many others).



Key specs

Speed	35.86 TFLOPS LINPACK (40 TF peak)
Processors	5,120 NEC SX-6 vector processors
Nodes	640 nodes, 8 vector CPUs each
Memory	10 TB total
Interconnect	Single-stage crossbar (IXS)
Power	~6 MW

<https://www.tomshardware.com/picturestory/696-supercomputer-evolution-over-22-years.html>

IBM Roadrunner (2008)



Los Alamos National Laboratory

Built by IBM for NNSA

When & where. The first machine to break one petaflop, in May 2008. About \$100 million. Used by the NNSA for **REDACTED**.

Architecture. A hybrid, heterogeneous design: AMD dual-core Opteron CPUs each paired with an IBM PowerXCell 8i accelerator (a chip descended from the PlayStation 3 Cell). The Cell chips did most of the math, the Opterons ran the system.

Network topology. InfiniBand in a fat-tree, organized as about 18 connected units of roughly 180 nodes each, 288-port InfiniBand switches and about 55 miles of optical cable.

Speed in perspective. One quadrillion operations per second. IBM's line at the time: it would take 100,000 of the fastest 2008 laptops, stacked 1.5 miles high, to match it.

Why it mattered. First petaflop machine, and the first to showcase accelerator-based heterogeneous computing. *Extremely* power-efficient.



Key specs

Speed	1.026 PFLOPS LINPACK (first petaflop)
Processors	~6,480 AMD Opteron + ~12,960 IBM Cell
Nodes	~3,240 TriBlade nodes
Memory	~104 TB
Interconnect	InfiniBand fat-tree
Power	2.35 MW

https://en.wikipedia.org/wiki/National_Nuclear_Security_Administration

<https://www.tomshardware.com/picturestory/696-supercomputer-evolution-over-22-years-2.html>

https://en.wikipedia.org/wiki/Roadrunner_%28supercomputer%29

IBM Roadrunner (2008)



Los Alamos
Built by IBM

When & where
2008. About \$

Architecture.
Opteron CPUs
chip descended
of the math, the

Network topology
connected units
switches and a

Speed in person
line at the time
stacked 1.5 miles

Why it matters
accelerator-based
power-efficient



urity Administration



etaflop)

0 IBM Cell

https://en.wikipedia.org/wiki/National_Nuclear_Security_Administration

<https://www.tomshardware.com/picturestory/696-supercomputer-evolution-over-22-years-2.html>

https://en.wikipedia.org/wiki/Roadrunner_%28supercomputer%29

Intrepid (2008)

IBM Blue Gene/P

Argonne Leadership Computing Facility (ALCF)

When & where. Argonne's first leadership Blue Gene. On the June 2008 TOP500 it ranked third overall and first in the world for open science. A workhorse for the DOE INCITE program.

Architecture. IBM Blue Gene/P: 40 racks, 40,960 nodes, each a quad-core PowerPC 450 at 850 MHz, for 163,840 cores and about 80 TB of memory. A deliberately low-power, many-core design.

Network topology. A 3-D torus for point-to-point messaging, plus separate collective (tree) and global-barrier networks.

Speed in perspective. About 450 trillion operations per second on LINPACK (557 TF peak). Modest next to that year's Roadrunner, but far more power-efficient, and the top open-science machine in the world.

Why it mattered. Put Argonne's ALCF on the map and made the energy-efficient Blue Gene line a mainstay for peer-reviewed open science. Seriously, people still talk about this machine with starry-eyes.



Key specs

Speed	450.3 TFLOPS LINPACK (557 TF peak)
Processors	40,960 quad-core PowerPC 450 (850 MHz)
Nodes	40,960 nodes, 163,840 cores
Memory	~80 TB
Interconnect	3-D torus (Blue Gene/P)
Power	~1.3 MW

<https://news.softpedia.com/news/DOE-Awards-Millions-of-Processing-Hours-149742.shtml>

Mira (2012)

IBM Blue Gene/Q

Argonne Leadership Computing Facility (ALCF)

When & where. Argonne's Blue Gene/Q, operational 2012. Debuted third on the June 2012 TOP500 and stayed a top open-science machine for years. Among the most energy-efficient of its era.

Architecture. IBM Blue Gene/Q: 48 racks, 49,152 nodes, each a 16-core PowerPC A2 at 1.6 GHz, for 786,432 cores and 768 TB of memory. About 20 times Intrepid's power.

Network topology. A 5-D torus interconnect (the first Blue Gene to use fiber optics for it), which lowers the average hop count between nodes.

Speed in perspective. About 8.6 quadrillion operations per second on LINPACK (10 PF peak). Its low-speed, high-core design made amazing at performance per watt.

Why it mattered. A mainstay of open science at ALCF, from cosmology and lattice QCD to materials and reactor design, and a high point for the energy-efficient Blue Gene approach. People love this machine. Pieces of it are kept around Argonne as decoration.



Key specs

Speed	8.59 PFLOPS LINPACK (10.06 PF peak)
Processors	49,152 x 16-core PowerPC A2 (1.6 GHz)
Nodes	49,152 nodes, 786,432 cores
Memory	768 TB
Interconnect	5-D torus (Blue Gene/Q)
Power	3.9 MW

[https://en.wikipedia.org/wiki/Mira_\(supercomputer\)](https://en.wikipedia.org/wiki/Mira_(supercomputer))

<https://www.cpushack.com/2013/02/17/ibm-blue-geneq-the-heart-of-a-supercomputer/>

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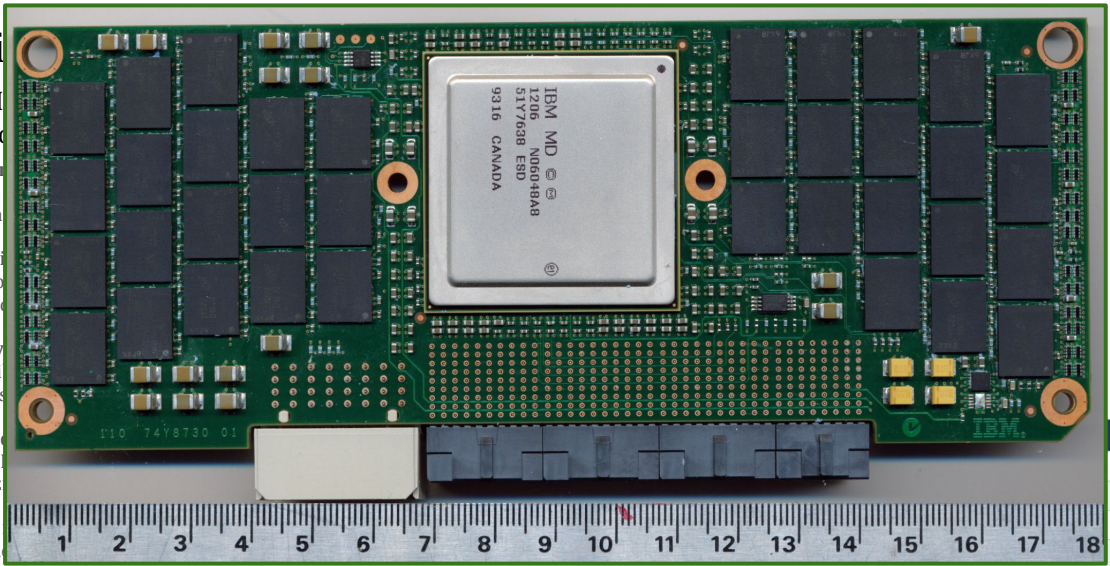
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Interconnect	5-D torus (Blue Gene/Q)
Power	3.9 MW

[https://en.wikipedia.org/wiki/Mira_\(supercomputer\)](https://en.wikipedia.org/wiki/Mira_(supercomputer))

<https://www.cpushack.com/2013/02/17/ibm-blue-geneq-the-heart-of-a-supercomputer/>

Cray Titan (2012)

Cray XK7

Oak Ridge National Laboratory, Tennessee

When & where. No. 1 from late 2012 to mid-2013. An upgrade of the earlier Jaguar machine. About \$97 million.

Architecture. 18,688 nodes, each a 16-core AMD Opteron 6274 paired with one NVIDIA Tesla K20X GPU. The first GPU-accelerated hybrid to exceed 10 petaflops.

Network topology. The Cray Gemini interconnect arranged as a 3-D torus.

Speed in perspective. ORNL says: on par with each of the world's roughly 7 billion people carrying out 3 million calculations every second, all at once.

Why it mattered. Brought GPUs into the mainstream at the leadership scale. Its pattern (CPU plus GPU nodes on a fast torus) led directly to Summit and Frontier, the current workhorses. Titan nodes are shipped around to Supercomputing (conference) for decor.



Key specs

Speed	8.59 PFLOPS LINPACK (10.06 PF peak)
Processors	49,152 x 16-core PowerPC A2 (1.6 GHz)
Nodes	49,152 nodes, 786,432 cores
Memory	768 TB
Interconnect	5-D torus (Blue Gene/Q)
Power	3.9 MW

[https://en.wikipedia.org/wiki/File:Oak_Ridge_-_Kraken_\(Cray_XT5\).JPG](https://en.wikipedia.org/wiki/File:Oak_Ridge_-_Kraken_(Cray_XT5).JPG)

[https://en.wikipedia.org/wiki/Jaguar_\(supercomputer\)](https://en.wikipedia.org/wiki/Jaguar_(supercomputer))

https://en.wikipedia.org/wiki/Titan_%28supercomputer%29

<https://wccfttech.com/nvidia-unveils-tesla-k20x-tesla-k20-worlds-fastest-efficient-accelerators/>

<https://www.ornl.gov/news/ornl-debuts-titan-supercomputer>

Cray Titan (2012)

Cray XK7
Oak Ridge National

When & where. No. 1 from the earlier Jaguar machine

Architecture. 18,688 nodes paired with one NVIDIA Tesla hybrid to exceed 10 petaflop

Network topology. The Cray torus.

Speed in perspective. ORNL roughly 7 billion people calculate second, all at once.

Why it mattered. Brought leadership scale. Its pattern led directly to Summit and nodes are shipped around the decor.



OPS LINPACK (10.06 PF peak)

16-core PowerPC A2 (1.6 GHz)

nodes, 786,432 cores

Interconnect	5-D torus (Blue Gene/Q)
Power	3.9 MW

[https://en.wikipedia.org/wiki/File:Oak_Ridge_-_Kraken_\(Cray_XT5\).JPG](https://en.wikipedia.org/wiki/File:Oak_Ridge_-_Kraken_(Cray_XT5).JPG)

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<https://www.ornl.gov/news/ornl-debuts-titan-supercomputer>

Cray Titan (2012)

Cray XK7

Oak Ridge National

When & where. No. 1 from the early 2010s

Archit

paired hybrid

Network torus.

Speed roughly second

Why it mattered. Brought leadership scale. Its pattern led directly to Summit and nodes are shipped around decor.



OPS LINPACK (10.06 PF peak)

16-core PowerPC A2 (1.6 GHz)

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<https://www.ornl.gov/news/ornl-debuts-titan-supercomputer>

Summit (2018)

IBM AC922

Oak Ridge National Laboratory

When & where. No. 1 from June 2018 to June 2020, returning the US to the top. About \$325 million with its sister machine Sierra.

Architecture. About 4,608 nodes, each with two IBM POWER9 CPUs (22 cores) and six NVIDIA Volta V100 GPUs, linked inside the node by NVLink, with over 600 GB of coherent memory per node and a 1.6 TB burst buffer.

Network topology. Dual-rail Mellanox EDR InfiniBand in a three-level, non-blocking fat tree, so any two nodes can talk at full bandwidth regardless of other traffic.

Speed in perspective. About 150 quadrillion operations per second. Reached multiple exaflops in mixed precision and 1.88 exaflops on a genomics run.

Why it mattered. A GPU-and-NVLink workhorse for both simulation and large-scale AI, and a bridge toward exascale.



Key specs	
Speed	148.6 PFLOPS LINPACK (200 PF peak)
Processors	9,216 IBM POWER9 + 27,648 NVIDIA V100 GPU
Nodes	~4,608 nodes
Memory	>10 PB total
Interconnect	Dual-rail EDR InfiniBand, fat-tree
Power	13 MW

[https://en.wikipedia.org/wiki/Summit_\(supercomputer\)](https://en.wikipedia.org/wiki/Summit_(supercomputer))

<https://www.tomshardware.com/news/summit-supercomputer-nvidia-ibm-power9-volta-35962.html>

Summit (2)

IBM AC922
Oak Ridge National Laboratory

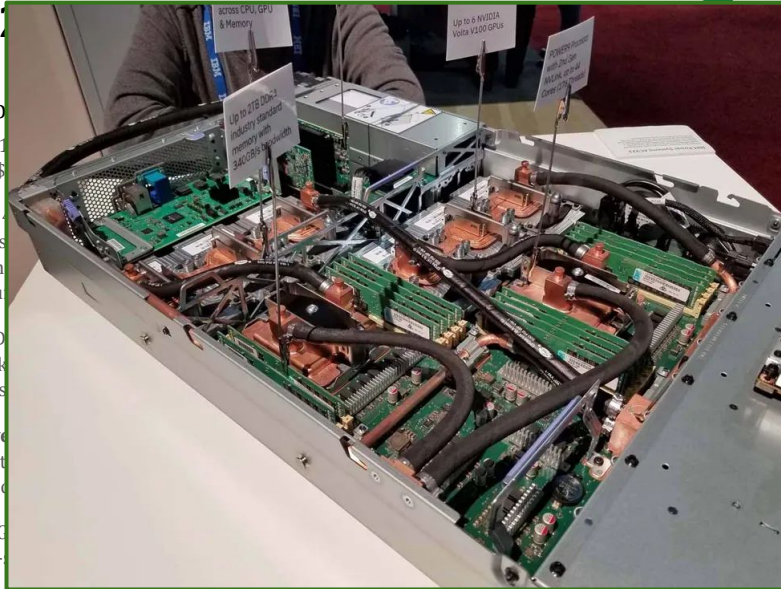
When & where. No. 1 in the US to the top. About \$1.3 billion

Architecture. About 4,000 CPUs (22 cores) and 56 nodes by NVLink, with 1.6 TB burst buffer

Network topology. D3, three-level, non-blocking, 100 Gbps bandwidth regardless of distance

Speed in perspective. 130 petaflops per second. Reached multiple exaflops on a genomic simulation

Why it mattered. A CERN simulation and large-scale



(200 PF peak)
27,648 NVIDIA V100 GPU
d, fat-tree

[https://en.wikipedia.org/wiki/Summit_\(supercomputer\)](https://en.wikipedia.org/wiki/Summit_(supercomputer))

https://www.tomshardware.com/news/summit-supercomputer-nvidia-ibm-power9-volta_35962.html

Summit (2

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Oak Ridge Nation

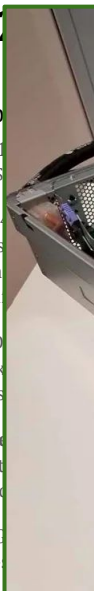
When & where. No. 1
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Architecture. About
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Network topology. D
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27,648 NVIDIA V100 GPU
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[https://en.wikipedia.org/wiki/Summit_\(supercomputer\)](https://en.wikipedia.org/wiki/Summit_(supercomputer))

https://www.tomshardware.com/news/summit-supercomputer-nvidia-ibm-power9-volta_35962.html

Summit

IBM AC920
Oak Ridge

When & where
US to the top

Architecture
CPUs (22 core)
node by NVL
and a 1.6 TB

Network topology
three-level, 100
bandwidth 100

Speed in peak
second. Reached
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Why it matters
simulation a



The image shows a red book titled "tom's HARDWARE" with the subtitle "A Purch Brand" and a grey NVIDIA V100 GPU. The GPU has a label with the text "NVP10000 9916 CR P" and a QR code.

V100 GPU

COMP 150, Spring

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[https://en.wikipedia.org/wiki/Summit_\(supercomputer\)](https://en.wikipedia.org/wiki/Summit_(supercomputer))

https://www.tomshardware.com/news/summit-supercomputer-nvidia-ibm-power9-volta_35962.html

Fugaku (2020)

IBM AC922

Oak Ridge National Laboratory

When & where. No. 1 from June 2020 to June 2022. Named for Mount Fuji. A roughly \$1 billion national program, deployed early to help with COVID-19 research.

Architecture. 158,976 nodes, each a single Fujitsu A64FX Arm CPU with 48 compute cores, 512-bit SVE vectors, and on-package HBM2 memory. No separate GPUs; the CPU itself is the vector engine.

Network topology. The Tofu Interconnect D, a 6-D mesh/torus in which each node links directly to ten neighbors, with hardware support for collective reductions.

Speed in perspective. About 440 quadrillion operations per second. Put to work early on COVID-19 problems such as droplet-spread modeling and drug screening.

Why it mattered. The first Arm-based machine to reach No. 1, and the first to top LINPACK, HPCG, Graph500, and HPL-AI at the same time.



Key specs	
Speed	442 PFLOPS LINPACK (537 PF peak)
Processors	158,976 Fujitsu A64FX Arm CPUs (48 cores)
Nodes	158,976 nodes
Memory	~4.85 PB HBM2
Interconnect	Tofu Interconnect D, 6-D torus
Power	~30 MW

[https://en.wikipedia.org/wiki/Fugaku_\(supercomputer\)](https://en.wikipedia.org/wiki/Fugaku_(supercomputer))

<https://www.bloomberg.com/news/articles/2020-06-23/fujitsu-s-arm-powered-supercomputer-claims-world-speed-crown>

<https://soranews24.com/2025/08/21/supercomputer-noodles-bring-the-magic-of-fugaku-to-the-dining-table/>

Fugaku (2020)

IBM AC922

Oak Ridge National Lab

When & where. No. 1 from June 2020, Mount Fuji. A roughly \$1 billion help with COVID-19 research.

Architecture. 158,976 nodes, each with 48 compute cores, 512-bit memory. No separate GPUs; the

Network topology. The Tofu Interconnect, which each node links directly to support for collective reduction

Speed in perspective. About 4 seconds. Put to work early on COVID droplet-spread modeling and d

Why it mattered. The first Arm-based the first to top LINPACK, HPCG, time.



Interconnect	Tofu Interconnect D, 6-D torus
Power	~30 MW

[https://en.wikipedia.org/wiki/Fugaku_\(supercomputer\)](https://en.wikipedia.org/wiki/Fugaku_(supercomputer))

<https://www.bloomberg.com/news/articles/2020-06-23/fujitsu-s-arm-powered-supercomputer-claims-world-speed-crown>

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Fugaku

IBM AC922
Oak Ridge National Laboratory

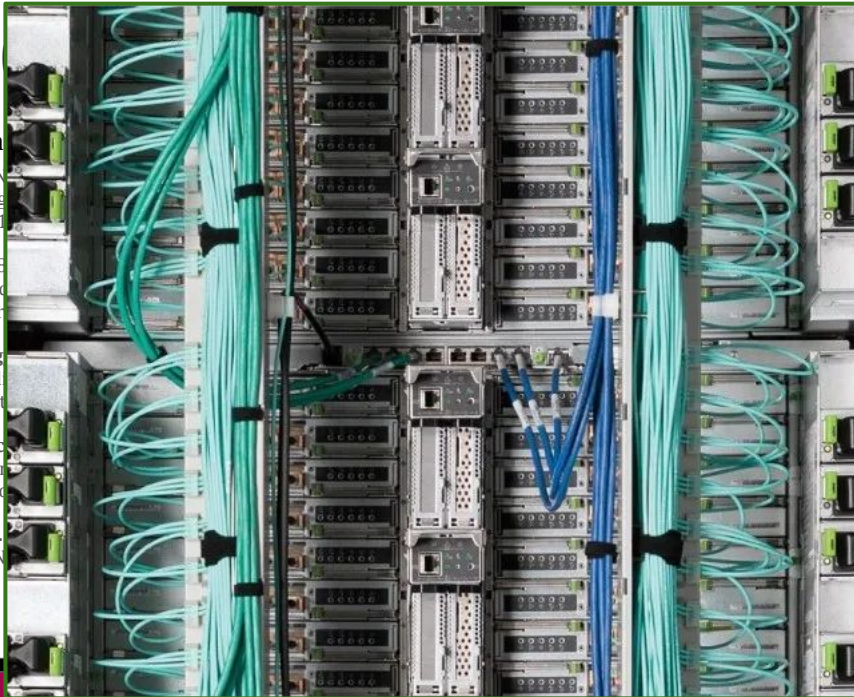
When & where. November 2020. Mount Fuji. A rough help with COVID-19.

Architecture. 158 nodes with 48 compute nodes per node. No separate memory.

Network topology. A mesh topology which each node has a support for collective communication.

Speed in perspective. 1.1 seconds. Put to work on droplet-spread model.

Why it mattered. The first to top LINPACK time.



peak)
PUs (48 cores)

[https://en.wikipedia.org/wiki/Fugaku_\(supercomputer\)](https://en.wikipedia.org/wiki/Fugaku_(supercomputer))

<https://www.bloomberg.com/news/articles/2020-06-23/fujitsu-s-arm-powered-supercomputer-claims-world-speed-crown>

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Frontier (2022)

HPE Cray EX
Oak Ridge National Laboratory

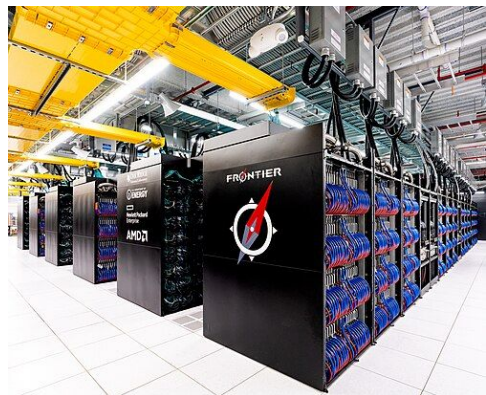
When & where. The first exascale machine, No. 1 from 2022. About \$600 million, 100% warm-water liquid cooled. Also topped the Green500 efficiency list at launch.

Architecture. About 9,400 nodes, each one AMD EPYC Trento 64-core CPU with four AMD Instinct MI250X GPUs, for over 37,000 GPUs and roughly 8.7 million total cores.

Network topology. HPE Slingshot-11 at 200 Gb/s in a dragonfly topology.

Speed in perspective. One quintillion (10^{18}) operations per second. If all 8 billion people on Earth each did one calculation per second, it would take them about four years to do what Frontier does in one second. At launch it beat the next seven TOP500 systems combined.

Why it mattered. Broke the exascale barrier and set the current template for the largest machines. In 2024 LLNL's El Capitan, also AMD-based, became the new No. 1.



Key specs

Speed	1.102 EFLOPS LINPACK (first exaflop)
Processors	~9,400 AMD EPYC + ~37,000 AMD MI250X GPU
Nodes	~9,400 nodes, ~8.7M cores
Memory	~9.2 PB
Interconnect	HPE Slingshot-11, dragonfly
Power	~21 to 25 MW

<https://www.flickr.com/photos/oakridgelab/52280656148/>

[https://en.wikipedia.org/wiki/Frontier_\(supercomputer\)](https://en.wikipedia.org/wiki/Frontier_(supercomputer))

<https://mindmatters.ai/2022/06/worlds-fastest-computer-breaks-into-the-exascale/>

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[https://en.wikipedia.org/wiki/Frontier_\(supercomputer\)](https://en.wikipedia.org/wiki/Frontier_(supercomputer))

<https://mindmatters.ai/2022/06/worlds-fastest-computer-breaks-into-the-exascale/>

Aurora (2024)

HPE Cray EX

Argonne Leadership Computing Facility (ALCF)

When & where. Argonne's exascale machine. Second computer to cross the exaflop line, after Frontier, and briefly the world's number two.

Architecture. HPE Cray EX: 10,624 blades, each with two Intel Xeon Max (Sapphire Rapids, with HBM) CPUs and six Intel Data Center GPU Max (Ponte Vecchio) accelerators, for 21,248 CPUs and 63,744 GPUs.

Network topology. HPE Slingshot-11 in a dragonfly topology with adaptive routing.

Speed in perspective. Over one quintillion operations per second on LINPACK (1.012 EF Rmax, about 2 EF peak). On the mixed-precision AI benchmark it reached 10.6 exaflops, first in the world.

Why it mattered. Argonne's flagship for the exascale era. Intel GPUs are a novel architecture, especially at this scale.



Key specs	
Speed	1.012 EFLOPS LINPACK (~2 EF peak)
Processors	21,248 Intel Xeon Max + 63,744 Intel GPU Max
Nodes	10,624 nodes
Memory	~20 PB DDR5 + ~10 PB HBM
Interconnect	HPE Slingshot-11, dragonfly
Power	38.7 MW

<https://www.hardwareluxx.de/index.php/news/allgemein/wirtschaft/63570-top500-liste-nur-ein-neuer-supercomputer-in-den-top-ten.html>

[https://en.wikipedia.org/wiki/Aurora_\(supercomputer\)](https://en.wikipedia.org/wiki/Aurora_(supercomputer))

<https://cloudnews.tech/aurora-intels-exascale-supercomputer-that-marks-a-turning-point-in-computing-history/>

<https://www.anl.gov/article/argonne-releases-aurora-exascale-supercomputer-to-researchers-enabling-accelerated-science>

<https://dataconomy.com/2024/07/08/aurora-supercomputer-worlds-fastest/>

Aurora

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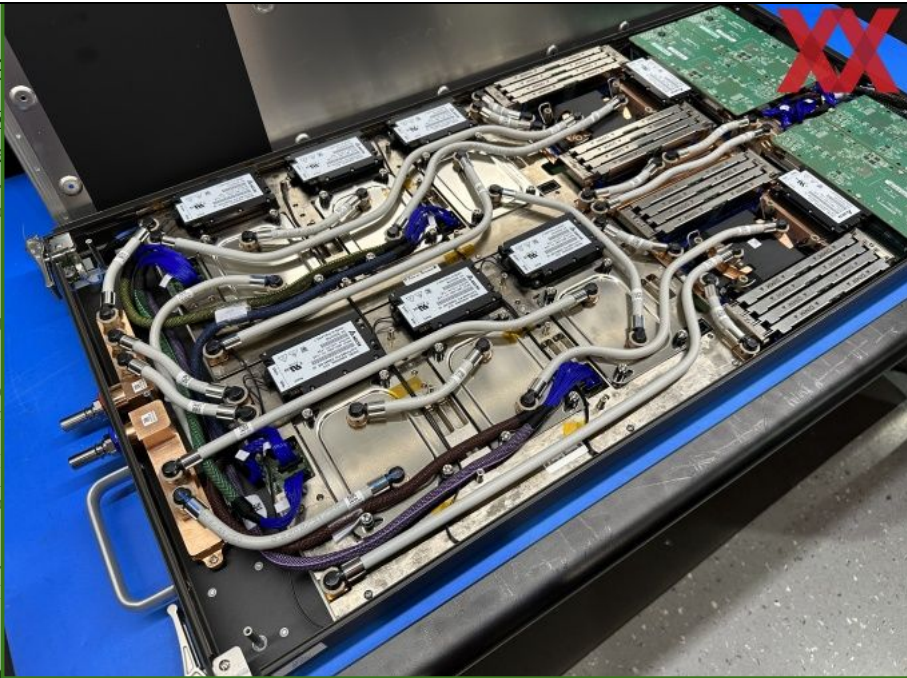
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<https://www.anl.gov/article/argonne-releases-aurora-exascale-supercomputer-to-researchers-enabling-accelerated-science>

<https://dataconomy.com/2024/07/08/aurora-supercomputer-worlds-fastest/>

Questions?

Project 1 (and 3.A): Surveys!

- What is a survey?
 - Wide coverage of a topic
 - Uses books, articles, and peer-reviewed materials to explore a subject space
- Why will we do them?
 - Standard in research
 - Honestly, kinda fun
- How will they be graded?
 - Read the rubric, and look at the example and associated rubric.
 - Cite your sources
 - Review our AI policy

Look for the rubric by EoD tomorrow, Sep 3rd.

Review

- Course flushed-out
- P1 out today or tomorrow, due 19 September 2026, 11:59PM AoE
 - Look for a rubric and assignment sheet
- Systems:
 - Taxonomy
 - Historic systems
 - Modern systems
 - Networks, architectures, operating systems (the beginnings...)
- Next week:
 - R01 and HW1 - posted by Friday
 - Networks, subjects, and more detail on compute architectures